SBMS | NAME SIMILARITY FEATURE COMPARISON

XGBoost with Jaro-Winkler vs normalized Levenshtein

Purpose and setup

Compared the name feature while keeping the XGBoost model and five other pair features fixed.
Training: 12,000 pairs (6,000 per class).
Holdout: 12,000 pairs (6,000 per class); group overlap: 0.
Both runs used five-fold grouped OOF evaluation, threshold 0.50, and identical model settings.

Metric	Jaro-Winkler	Levenshtein
Grouped OOF F1	**90.67%**	90.19%
Holdout accuracy	**90.77%**	90.12%
Holdout precision	92.04%	**92.32%**
Holdout recall	**89.27%**	87.52%
Holdout F1	**90.63%**	89.85%
Holdout ROC-AUC	**96.85%**	96.31%
Holdout PR-AUC	**97.27%**	96.88%
False positives	**463**	437
False negatives	644	**749**

Interpretation

Jaro-Winkler led on grouped OOF F1 and most holdout metrics. It made 1,107 total holdout errors versus 1,186 for Levenshtein. Levenshtein had 26 fewer false positives, but 105 more false negatives. Jaro-Winkler remains the selected name feature for the XGBoost model.

Limitations

These are synthetic pairs, not real beneficiary records. The holdout was previously used in the Random Forest vs XGBoost comparison, so it is not a pristine first-use evaluation. The grouped OOF comparison is the cleaner evidence here; real-world accuracy remains unknown.

Source: ml-service/evaluation/xgboost_name_similarity_comparison.json